Blueprint Workflow & Fitur Role Admin — CafeNow

Teknologi: Laravel (PHP 8.4, `declare(strict_types=1)`) + Eloquent ORM + Sanctum + Inertia/React (TypeScript) **Persistence:** SQL Database (MySQL, Eloquent ORM) **Arsitektur:** Layered (Route → Controller → Service → Model), tanpa repository layer **Entry Point:** API (JSON) + halaman Inertia untuk rendering SPA **Cakupan dokumen:** seluruh endpoint role admin (`routes/api/admin.php`, 25 endpoint) + halaman admin + auth admin.

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1. Daftar Endpoint Admin

Semua API admin berada di balik `middleware(['auth:sanctum', 'role:admin'])` dengan prefix `/api/admin` (`routes/api/admin.php:17`). Pengecualian: login admin di `POST /api/admin/login` (public, `throttle:auth`).

Grup	Method	URI	Controller@Method
Auth	POST	`/api/admin/login`	`Auth\AdminAuthController@login`
Dashboard	GET	`/api/admin/dashboard`	`AdminDashboardController@index`
Dashboard	GET	`/api/admin/dashboard/recent-activity`	`AdminDashboardController@recentActivity`
Analytics	GET	`/api/admin/analytics`	`AdminAnalyticsController@index` (render Inertia)
Analytics	GET	`/api/admin/analytics/platform`	`@platform`
Analytics	GET	`/api/admin/analytics/revenue`	`@revenue`
Analytics	GET	`/api/admin/analytics/cafes`	`@topCafes`
Analytics	GET	`/api/admin/analytics/bookings/status`	`@bookingStatus`
Analytics	GET	`/api/admin/analytics/commission`	`@commission`
Analytics	GET	`/api/admin/analytics/export-csv`	`@exportCsv`
Cafes	GET	`/api/admin/cafes`	`AdminCafeController@index`
Cafes	GET	`/api/admin/cafes/{cafe}`	`@show`
Cafes	POST	`/api/admin/cafes/{cafe}/suspend`	`@suspend`
Cafes	POST	`/api/admin/cafes/{cafe}/unsuspend`	`@unsuspend`
Approvals	GET	`/api/admin/cafe-approvals`	`AdminCafeRegistrationController@index`
Approvals	GET	`/api/admin/cafe-approvals/{cafe}`	`@show`
Approvals	POST	`/api/admin/cafe-approvals/{cafe}/approve`	`@approve`
Approvals	POST	`/api/admin/cafe-approvals/{cafe}/reject`	`@reject`
Featured	GET/POST/PUT/DELETE	`/api/admin/featured-cafes` + `/{featuredCafe}`	`AdminFeaturedCafeController` (`apiResource` minus `show`)

Grup	Method	URI	Controller@Method
Bookings	GET	`/api/admin/bookings`	`AdminBookingController@index`
Bookings	GET	`/api/admin/bookings/{booking}`	`@show`
Bookings	POST	`/api/admin/bookings/{booking}/cancel`	`@cancel`
Refunds	GET	`/api/admin/refunds`	`AdminRefundController@index`
Refunds	GET	`/api/admin/refunds/{refund}`	`@show`
Refunds	POST	`/api/admin/refunds/{refund}/process`	`@process`
Refunds	POST	`/api/admin/refunds/{refund}/reject`	`@reject`
Commissions	GET	`/api/admin/commissions/rates`	`AdminCommissionController@rates`
Commissions	GET	`/api/admin/commissions/history`	`@history`
Commissions	PATCH	`/api/admin/commissions/rates/{type}`	`@updateRate`
Users	GET	`/api/admin/users`	`AdminUserController@index`
Users	PATCH	`/api/admin/users/{user}`	`@update`
Users	POST	`/api/admin/users/{user}/suspend`	`@suspend`
Users	POST	`/api/admin/users/{user}/unsuspend`	`@unsuspend`
Audit	GET	`/api/admin/audit-logs`	`AdminAuditLogController@index`

Halaman Inertia admin (`routes/web.php:98-110` , juga `role:admin`): `admin/dashboard` , `admin/cafe-approvals` , `admin/cafes` , `admin/featured-cafes` , `admin/bookings` , `admin/orders` , `admin/refunds` , `admin/analytics` , `admin/commissions` , `admin/users` , `admin/audit-logs` — dirender dari `resources/js/pages/admin/**` .

> **Catatan temuan:** `AdminOrderController` (`app/Http/Controllers/Admin/AdminOrderController.php`) ada dan lengkap (list order dengan `MenuOrderResource`) tetapi **TIDAK terdaftar** di `routes/api/admin.php` . Halaman `admin/Order/Oversight` terdaftar tapi tidak punya sumber data API. Ini dead code — endpoint `GET /api/admin/orders` perlu didaftarkan atau controller dihapus.

2. Workflow Per Grup Endpoint

2.1 Auth — Login Admin

Tujuan: otentikasi admin untuk mendapatkan token Sanctum dengan kemampuan `role:admin` . **Trigger:** submit form login di halaman admin. **File:** `AdminAuthController.php` , `LoginRequest.php` , `User.php` , `UserResource.php` .

Alur: 1. `POST /api/admin/login` (throttled `auth`) → `AdminAuthController@login` 2. `LoginRequest` memvalidasi `email` (required, email) & `password` (required) 3. `User::where('email', ...)->first()` + `Hash::check()` 4. Guard: `$user->role !== UserRole::Admin` → 403 `FORBIDDEN` 5. `createToken('admin-token', ['role:admin'])` → token + `UserResource`

Poin penting: gagal kredensial dilempar sebagai `ValidationException::withMessages(['email' => ...])` → 422 dengan `{errors: {email: [...]}}` . Bukan wrapper `success:false` (inkonsisten dengan controller lain).

```
$token = $user->createToken('admin-token', ['role:admin'])->plainTextToken;
```

Guard role global (`CheckRole.php`): `abort(403, 'Tidak memiliki akses')` jika role tidak cocok — diterapkan di route group, bukan per-method.

2.2 Dashboard

Tujuan: ringkasan metrik platform + aktivitas terbaru admin. **Trigger:** admin membuka `/admin/dashboard` . **File:** `AdminDashboardController.php` , `CommissionService.php` , `AdminAuditLog.php` .

`index()` — render Inertia `admin/dashboard` ; metrik dihitung via query Eloquent langsung: - `totalCafes` : count `approval_status = active` - `revenueToday` : `sum(total_price)` booking completed hari ini × `activeBookingRate()/100` (rate komisi aktif) - `activeBookings` : count status `pending` - `pendingApprovals` : count cafe `pending` - `newCustomers` : user role customer dibuat hari ini - `newOrders` : booking dibuat hari ini

`recentActivity()` — 10 log terakhir `AdminAuditLog` (dengan relasi `admin`), diterjemahkan ke label bahasa Indonesia via `activityMessage()` (map action → label) dan `entityLabel()` (resolve nama entitas: cafe→nama, user→nama, booking→referensi, refund→ `#id` , featured→nama cafe).

2.3 Analytics

Tujuan: laporan bisnis + ekspor CSV. **File:** `AdminAnalyticsController.php` , `Booking.php` , `Cafe.php` , `User.php` .

Endpoint	Logika
`platform`	Count per role user, per status cafe, per status booking; `sum(cafe_receives)` booking completed
`revenue?start_date&end_date`	Group-by `DATE(created_at)` , `COALESCE(SUM(commission),0)` booking completed
`topCafes?top=N`	`withCount('bookings')` + `withSum('bookings','commission')` , filter `status='approved'` (kolom legacy — beda konvensi, lihat Pitfall), sort desc, limit N
`bookingStatus`	Count completed / pending / cancelled
`commission?period=daily|weekly|monthly&from&to`	Validasi period; `whereBetween('booking_date')` ; group-by format `Y-m-d` / `Y-W` / `Y-m` ; sum commission, total_price, count
`exportCsv?start_date&end_date`	CSV manual: header `Tanggal,Cafe,Revenue` , baris per booking completed; response `text/csv` + `Content-Disposition: attachment`

2.4 Manajemen Cafe (List, Suspend, Unsuspend)

Tujuan: pencarian & moderasi cafe aktif. **Trigger:** aksi admin di halaman `admin/Cafe/Management` . **File:** `AdminCafeController.php` , `BookingService.php` , `Cafe.php` , `Booking.php` .

`index` : filter `keyword` (LIKE nama cafe **atau** nama owner via `orWhereHas`), filter `status` (approval_status), paginate default 25.

`suspend` — alur ber-transaksi:

```
DB::transaction(function () use ($cafe): void {
    $cafe->transitionTo(CafeApprovalStatus::Suspended);

    $unpaid = $cafe->bookings()
        ->where('payment_status', 'unpaid')
        ->whereIn('status', ['pending', 'confirmed'])
        ->get();

    foreach ($unpaid as $booking) {
        $this->bookingService->cancel(
            $booking,
            CancelledBy::System->value,
            'Dibatalkan otomatis karena cafe di-suspend',
        );
    }
});
```

- Transisi status via state machine `Cafe::transitionTo()` → exception `InvalidCafeStatusTransitionException` jika ilegal (mis. active→rejected)
- **Kompensasi (compensating action):** semua booking unpaid yang pending/confirmed dibatalkan otomatis dengan alasan sistem (dalam 1 transaksi DB)

`unsuspend` : guard manual `approval_status !== suspended` → `InvalidCafeStatusTransitionException` ; lalu `transitionTo(Active)` .

Keduanya menulis audit log via trait `LogsAdminActions` (`AdminAction::SuspendCafe` / `UnsuspendCafe`).

2.5 Persetujuan Cafe (Approvals)

Tujuan: review registrasi cafe oleh owner; setujui/tolak. **Trigger:** aksi di halaman `admin/Cafe/Approvals` . **File:** `AdminCafeRegistrationController.php` , `CafeApprovalStatus.php` , `CafeResource.php` , `Notification.php` .

`index` : default filter `approval_status = pending` (kecuali query `status` diberikan), eager-load `owner, photos, tables, menuItems` , format manual via `formatCafe()` (bukan Resource — menyertakan foto/tabel/menu dengan URL `asset('storage/...')`).

`approve` : 1. `transitionTo(CafeApprovalStatus::Active)` + `update(['rejection_reason' => null])` 2. `AdminAuditLog::create` (action `approve_cafe` , before/after value) 3. `Notification::create` ke owner (`cafe_approved` , delivery_status `sent`)

`reject` : 1. Validasi inline `reason` required max 1000 (ada `RejectCafeRequest` di `app/Http/Requests/Admin/` tapi **tidak dipakai** — validasi inline di controller) 2. `transitionTo(Rejected)` + simpan `rejection_reason` 3. Audit log `reject_cafe` + notifikasi `cafe_rejected` berisi alasan

State machine `CafeApprovalStatus` :

```
pending   → active | rejected
rejected  → pending   (fix-then-resubmit)
active    → suspended
suspended → active
```

`Cafe::isContentEditable()` = true saat active/approved/rejected (owner bisa edit & resubmit).

2.6 Featured Cafes

Tujuan: kurasi cafe unggulan (tampil di halaman publik dengan urutan). **Trigger:** CRUD di halaman `admin/Cafe/Featured`. **File:** `AdminFeaturedCafeController.php`, `FeaturedCafe.php`, `FeaturedCafeResource.php`.

Method	Perilaku
`index`	Semua featured + relasi `cafe`, order `display_order`
`store`	Validasi `cafe_id` exists, `display_order` min 0, `active_from` required date, `active_to` after active_from; **guard duplikat** → 409 `CONFLICT`; `created_by = user()->id`; audit `add_featured_cafe`; 201
`update`	Validasi parsial; update 3 field; audit `update_featured_cafe`
`destroy`	Hard delete; audit `remove_featured_cafe`; 204 tanpa body

Catatan: controller ini (dan `AdminCafeRegistrationController`) memakai pola audit langsung `AdminAuditLog::create` dengan action **string literal** — bukan enum `AdminAction` seperti controller lain (inkonsistensi konvensi, lihat Pitfall).

2.7 Oversight Booking

Tujuan: pantau & batalkan booking dari sisi platform. **Trigger:** aksi di halaman `admin/Booking/Oversight`. **File:** `AdminBookingController.php`, `BookingService.php`, `CancelledBy.php`.

`index`: filter `keyword` (LIKE `booking_code`), `status`, `cafe_id`; paginate 50; eager-load `cafe, customer`.

`cancel`: 1. Guard status: hanya `pending` / `confirmed` (const `CANCELLABLE_STATUSES`) → selain itu 422 `INVALID_STATUS` 2. `bookingService->cancel($booking, CancelledBy::Admin->value, $reason)` — service: update status `cancelled` + `cancelled_by` + reason, **release lock tabel** (`LockService::releaseForBooking`), dispatch event `BookingCancelled` 3. Audit log `cancel_booking`

(Untuk `BookingService::cancel` via admin: guard waktu 3 jam hanya berlaku bila `cancelledBy === 'customer'` — admin dapat membatalkan kapan saja.)

2.8 Refund

Tujuan: proses permintaan refund dari customer. **Trigger:** aksi di halaman `admin/Refund/Queue`. **File:** `AdminRefundController.php`, `RefundRequest.php`.

Method	Perilaku
`index`	Filter `status`; paginate 50; eager-load `booking, customer`

Method	Perilaku
`show`	Detail refund + relasi
`process`	`update(status='completed', admin_notes)` ; audit `process_refund`
`reject`	`update(status='rejected', admin_notes=reason)` ; audit `reject_refund`

Catatan: tidak ada guard status (refund yang sudah completed bisa diproses lagi), tidak ada transaksi/validasi request, tidak ada pemanggilan `MidtransService::refund()` untuk refund aktual di sisi payment gateway. Workflow paling sederhana dari semua grup admin.

2.9 Komisi

Tujuan: kelola rate komisi platform (booking & menu_order). **Trigger:** aksi di halaman `admin/Commission/Management` . **File:** `AdminCommissionController.php` , `CommissionRule.php` , `CommissionService.php` .

Method	Perilaku
`rates`	Semua rule → `[{type, percentage}]`
`history`	Filter `type` ; paginate 50; eager-load `setByAdmin` ; latest first
`updateRate`	`CommissionRule::where('commission_type', $type)->first()` ; jika ada → update `percentage` (`$request->float('percentage')`), audit `update_commission` . **Tidak ada validasi** (range, type valid) dan tidak ada logika pembuatan rule baru

Konsumsi: `CommissionService::activeBookingRate()` (dipakai dashboard & perhitungan booking) membaca rule aktif (`active_from <= now` , `active_to >= now` atau null, latest `active_from`), fallback default 15% booking / 10% menu. `CommissionRule` berisi `set_by_admin_id` tapi `updateRate` **tidak mengisinya**.

2.10 Manajemen User

Tujuan: kelola user platform (nama/email, suspend). **Trigger:** aksi di halaman `admin/User/Management` . **File:** `AdminUserController.php` , `User.php` , `UserRole.php` .

`index` : filter `keyword` (name/email LIKE), `role` ; paginate 50.

`update` : `$request->only(['name', 'email'])` (tanpa FormRequest); audit `update_user` .

`suspend` — dua guard bisnis: 1. **CANNOT_SELF_SUSPEND** (422): `$user->id === $request->user()->id` 2. **LAST_ADMIN** (422): jika target admin aktif dan jumlah admin aktif ≤ 1 3. `update(['is_suspended' => true])` ; audit `suspend_user`

`unsuspend` : set false; audit `unsuspend_user` .

Pola error: `{success: false, error: {code, message}}` dengan status 422.

2.11 Audit Logs

Tujuan: jejak audit seluruh aksi admin (list, filter, paginate). **Trigger:** halaman `admin/User/AuditLogs`. **File:** `AdminAuditLogController.php`, `AdminAuditLog.php`.

Filter: `admin_id`, `action`, `entity_type`, `date_from` (`whereDate >=`), `date_to` (`whereDate <=`); paginate 100; eager-load `admin`; latest first.

Model **`AdminAuditLog`**: `timestamps = false`, kolom `admin_id, action, entity_type, entity_id, before_value (array), after_value (array), ip_address, created_at`. Semua mutasi status admin menulis 1 baris: action (enum/string), snapshot before/after, IP.

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3. Service Layer

Tidak ada service khusus admin (`app/Services/Admin` kosong). Controller admin memakai dua service domain:

`BookingService::cancel(Booking $booking, string $cancelledBy, ?string $reason = null): void` (`app/Services/Booking/BookingService.php:105`) - Guard 3 jam hanya untuk `cancelledBy === 'customer'` - Update status + `cancelled_by` + reason - `lockService->releaseForBooking($booking)` — lepaskan kunci slot - `BookingCancelled::dispatch($booking->id, $cancelledBy, $reason)` — event

`CommissionService` (`app/Services/Payment/CommissionService.php`) — `readonly class`, dipakai dashboard & perhitungan komisi booking: - `calculateBookingCommission(float $amount): float` - `activeBookingRate(): float` - `calculateMenuCommission(float $amount): float` - `getActiveRate(string $type)` — query `CommissionRule` aktif + fallback default

Pola DI: constructor injection `private readonly` (Laravel auto-resolve, tanpa binding manual). `AdminDashboardController` ← `CommissionService`; `AdminCafeController` & `AdminBookingController` ← `BookingService`.

Trait **`LogsAdminActions`** (`app/Traits/LogsAdminActions.php`) — cross-cutting concern audit:

```
protected function logAdminAction(
    Request $request, string $action, string $entityType,
    int $entityId, array $before = [], array $after = [],
): void
```

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4. Data Mapping Patterns

Tidak ada mapper DTO (tidak ada AutoMapper). Pola yang ada:

1. **Manual array mapping** — `AdminCafeRegistrationController::formatCafe()` membangun array response manual dengan denormalisasi (URL foto absolut, tabel/menu flatten, `location = city ?: address`).
2. **API Resource** — `CafeResource`, `FeaturedCafeResource`, `MenuOrderResource` untuk response terstruktur (approve/reject cafe, featured, order).
3. **Eloquent** **`only()`** — snapshot before/after untuk audit (`$user->only(['name','email'])`).
4. **Eloquent cast** — `CommissionRule.percentage` & `Booking.commission` di-cast `decimal:2`, `facilities` / `operational_hours` → `array`; serialisasi JSON otomatis.

5. Data Access Implementation

Pola: query langsung Eloquent di controller (tanpa repository). Semua query memakai query builder + eager loading + pagination:

```
$query = Cafe::with('owner');
if ($request->filled('keyword')) {
    $query->where('name', 'like', "%{$request->keyword}%")
          ->orWhereHas('owner', fn ($q) => $q->where('name', 'like', "%{$request->keyword}%"));
}
$cafes = $query->paginate($request->integer('per_page', 25));
```

Pola umum per list endpoint: - Filter opsional: `$request->filled('x')` + `where` - `$request->integer('per_page', 25|50|100)` — default bervariasi per controller - Response pagination: `{total, per_page, current_page, last_page}` (di sebagian controller diekstrak ke `paginationMeta()` privat)

Transaksi: hanya di `AdminCafeController@suspend` (`DB::transaction`) — karena multi-tulis (transisi status + pembatalan booking).

Relasi penting: `Cafe.owner/tables/menuItems/photos/bookings/featured` , `Booking.cafe/customer/table/payment/refundRequest` , `RefundRequest.booking/customer` , `FeaturedCafe.cafe/createdBy` , `AdminAuditLog.admin` , `CommissionRule.setByAdmin` .

6. Response Construction

Dua format response yang berbeda di codebase:

A. Format legacy (mayoritas):

```
{ "success": true, "data": [...], "pagination": { "total": ..., "per_page": ..., "current_page": ..., "last_page":
```

dipakai: cafes, bookings, refunds, commissions, users, audit-logs, dashboard recent-activity.

B. Format Resource (lebih baru):

```
{ "success": true, "data": [ResourceCollection] | Resource, "meta": { "current_page", "last_page", "total" } }
```

dipakai: `AdminFeaturedCafeController` , `AdminCafeRegistrationController` (approve/reject), `AdminOrderController` .

Status code: 200 default; **201** saat create (`featured-cafes` store); **204** delete; **422** validasi/bisnis guard (`INVALID_STATUS` , `CANNOT_SELF_SUSPEND` , `LAST_ADMIN`); **403** role/salah akun; **409** duplikat (`CONFLICT`); **500** exception tak tertangani.

Error response pattern:

```
{ "success": false, "data": null, "error": { "code": "CANNOT_SELF_SUSPEND", "message": "..." } }
```

(untuk `ValidationException` : `{errors: {field: [...]}}` — bentuk berbeda).

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7. Error Handling Patterns

Layer	Mekanisme
Route guard	`CheckRole` middleware → `abort(403)`
Validasi input	Inline `$request->validate([...])` mayoritas; `FormRequest` hanya di auth (`LoginRequest`) & reject cafe (`RejectCafeRequest` — tidak terpakai)
Bisnis guard	Return JSON 422 inline (booking cancel, user suspend) ATAU throw domain exception (`InvalidCafeStatusTransitionException`)
Global	Handler Laravel default; `ValidationException` → 422 `{errors}`

Catatan inkonsistensi (potensi bug): - Guard bisnis memakai dua gaya: throw exception (`Cafe::transitionTo`) vs return JSON 422 (booking cancel, user suspend). Exception `InvalidCafeStatusTransitionException` akan menghasilkan 500, bukan 422, karena tidak ada handler khusus → user melihat error generic. - `AdminCommissionController@updateRate` dan `AdminRefundController@process/reject` tidak punya validasi input maupun guard status. - `AdminAnalyticsController@topCafes` memakai `where('status', 'approved')` — kolom legacy; konsisten dengan `Cafe::transitionTo` yang memetakan `'approved'` → `Active` , tapi tidak konsisten dengan controller lain yang memakai `approval_status` .

Logging: tidak ada `Log::` eksplisit di controller admin; jejak mutasi tersimpan di `AdminAuditLog` (before/after + IP). Event `BookingCancelled` di-subscribe listener async (log + notifikasi).

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8. Asynchronous Processing Patterns

- **Event → Listener:** `BookingService::cancel` me-dispatch `BookingCancelled` ; `BookingConfirmed` dipakai flow lain. Tidak ada queue worker khusus — event listener sinkron.
- **Notifikasi in-DB:** `Notification::create` langsung (bukan `Notification` Laravel class) — tabel `notifications` diisi dengan `delivery_status = 'sent'` (approve/reject cafe).
- **Tidak ada** job queue, webhook, atau cron khusus admin. `exportCsv` sinkron (cocok untuk skala data kecil).

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9. Testing Approach

Test admin berada di `tests/Feature/Admin/` (Pest/PHPUnit, `RefreshDatabase`):

File	Cakupan
`AdminAnalyticsConsolidationTest.php`	Endpoint analytics + agregasi

File	Cakupan
`AdminAuditLoggingTest.php`	Penulisan audit log pada aksi admin
`AdminBookingControllerTest.php`	List & cancel booking
`AdminRefundControllerTest.php`	List, show, process, reject refund
`AdminUserSuspendGuardTest.php`	Guard `CANNOT_SELF_SUSPEND` & `LAST_ADMIN`
`DashboardCommissionTest.php`	Metrik dashboard + rate komisi

Pola fixture: `User::factory()->create(['role' => 'admin'])` lalu `$this->actingAs($admin)`, akses `getJson('/api/admin/...')` + assertion `assertJsonPath`. Contoh (`AdminRefundControllerTest`):

```
private function makeRefund(): array
{
    $admin = User::factory()->create(['role' => 'admin']);
    $customer = User::factory()->create(['role' => 'customer', 'name' => 'Budi Refund']);
    $booking = Booking::factory()->create(['customer_id' => $customer->id, 'payment_status' => 'paid']);
    $refund = RefundRequest::create([...]);
    return [$admin, $refund];
}
```

Belum ada test untuk: featured cafes, commissions, audit-log filter, cafe approvals, login admin, admin orders.

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10. Sequence Diagram

10.1 Login Admin

```
sequenceDiagram
    participant FE as Frontend (React)
    participant C as AdminAuthController@login
    participant DB as MySQL
    FE->>C: POST /api/admin/login {email, password}
    C->>C: LoginRequest validate
    C->>DB: User::where(email)->first()
    alt user null / Hash::check gagal
        C-->>FE: 422 ValidationException {errors.email}
    else role != admin
        C-->>FE: 403 {code: FORBIDDEN}
    else sukses
        C->>C: createToken('admin-token', ['role:admin'])
        C-->>FE: 200 {token, user: UserResource}
    end
```

10.2 Suspend Cafe (dengan kompensasi booking)

```
sequenceDiagram
    participant FE as Frontend
    participant C as AdminCafeController@suspend
```

```
    participant S as BookingService
    participant DB as MySQL
    FE->>C: POST /api/admin/cafes/{id}/suspend
    C->>DB: begin transaction
    C->>DB: Cafe::transitionTo(Suspended)
    alt transisi ilegal
        C-->>FE: 500 InvalidCafeStatusTransitionException
    end
    C->>DB: bookings unpaid & pending/confirmed
    loop setiap booking unpaid
        C->>S: cancel(booking, system, alasan)
        S->>DB: update status=cancelled
        S->>DB: releaseForBooking(booking)
        S->>DB: dispatch BookingCancelled (listener)
    end
    C->>DB: commit
    C->>DB: AdminAuditLog::create(suspend_cafe)
    C-->>FE: 200 {success: true}
```

10.3 Approve Cafe

```
sequenceDiagram
    participant FE as Frontend
    participant C as AdminCafeRegistrationController@approve
    participant DB as MySQL
    FE->>C: POST /api/admin/cafe-approvals/{id}/approve
    C->>DB: Cafe::transitionTo(Active)
    C->>DB: update rejection_reason = null
    C->>DB: AdminAuditLog::create(approve_cafe)
    C->>DB: Notification::create → owner (cafe_approved)
    C-->>FE: 200 {data: CafeResource}
```

10.4 Cancel Booking (Admin)

```
sequenceDiagram
    participant FE as Frontend
    participant C as AdminBookingController@cancel
    participant S as BookingService
    participant DB as MySQL
    FE->>C: POST /api/admin/bookings/{id}/cancel
    alt status bukan pending/confirmed
        C-->>FE: 422 {code: INVALID_STATUS}
    end
    C->>S: cancel(booking, admin, reason)
    S->>DB: update status=cancelled, cancelled_by, reason
    S->>DB: releaseForBooking (unlock slot)
    S->>DB: dispatch BookingCancelled
    C->>DB: AdminAuditLog::create(cancel_booking)
    C-->>FE: 200 {success: true}
```

10.5 Suspend User (dengan guard bisnis)

```
sequenceDiagram
    participant FE as Frontend
    participant C as AdminUserController@suspend
    participant DB as MySQL
    FE->>C: POST /api/admin/users/{id}/suspend
    alt user = diri sendiri
        C-->>FE: 422 {code: CANNOT_SELF_SUSPEND}
    else target admin & jumlah admin aktif <= 1
        C-->>FE: 422 {code: LAST_ADMIN}
    else valid
        C->>DB: update is_suspended = true
        C->>DB: AdminAuditLog::create(suspend_user)
        C-->>FE: 200 {success: true}
    end
```

11. Naming Conventions

Aspek	Konvensi	Contoh
Controller	`Admin{Entity}Controller` di `App\Http\Controllers\Admin`	`AdminRefundController`
Auth controller	`AdminAuthController` di `App\Http\Controllers\Auth` (bukan folder Admin)	—
Method	`index`, `show`, `store`, `update`, `destroy` (CRUD) + `{action}` verb untuk mutasi	`suspend`, `unsuspend`, `approve`, `reject`, `process`, `cancel`
Route naming	Prefix `/api/admin/{entity}`, action verb sebagai sub-route `POST .../suspend`	—
Trait audit	`LogsAdminActions` (bukan `AdminLoggingService`)	—
Enum	`AdminAction` (case PascalCase, value snake_case string)	`SuspendCafe = 'suspend_cafe'`
State machine	`CafeApprovalStatus` + method `transitionTo()` di model	—
Request	`{Entity}Request` di `App\Http\Requests\Admin`	`RejectCafeRequest`
Resource	`{Entity}Resource` di `App\Http\Resources`	`FeaturedCafeResource`
Service	`{Domain}Service` di `App\Services\{Domain}`	`BookingService`, `CommissionService`
Model	Singular PascalCase	`RefundRequest`, `CommissionRule`
File organization	`resources/js/pages/admin/{Domain}/{Feature}.tsx`	`admin/Refund/Queue.tsx`

Ketidakkonsistenan yang ditemukan: audit log ditulis dengan 2 gaya (enum `AdminAction` di sebagian, string literal di `AdminFeaturedCafeController` & `AdminCafeRegistrationController`); pagination meta berbentuk `pagination` di sebagian dan `meta` di sebagian lain; `AdminOrderController` tidak terdaftar di route.

12. Implementation Templates

12.1 Endpoint admin baru (mengikuti pola terbaik codebase)

1. Route (`routes/api/admin.php` — di dalam group `auth:sanctum` + `role:admin`):

```
Route::prefix('vouchers')->group(function () {
    Route::get('/', [AdminVoucherController::class, 'index']);
    Route::post('/{voucher}/activate', [AdminVoucherController::class, 'activate']);
});
```

2. Controller (`app/Http/Controllers/Admin/AdminVoucherController.php`) — pola: strict types, `readonly` DI, `LogsAdminActions` , guard inline JSON:

```
<?php

declare(strict_types=1);

namespace App\Http\Controllers\Admin;

use App\Enums\AdminAction;
use App\Http\Controllers\Controller;
use App\Models\Voucher;
use App\Traits\LogsAdminActions;
use Illuminate\Http\Request;
use Illuminate\Http\JsonResponse;

class AdminVoucherController extends Controller
{
    use LogsAdminActions;

    public function index(Request $request): JsonResponse
    {
        $query = Voucher::with('creator');

        if ($request->filled('status')) {
            $query->where('status', $request->string('status'));
        }

        $vouchers = $query->latest()->paginate($request->integer('per_page', 50));

        return response()->json([
            'success' => true,
            'data' => $vouchers->items(),
            'pagination' => [
                'total' => $vouchers->total(),
                'per_page' => $vouchers->perPage(),
                'current_page' => $vouchers->currentPage(),
                'last_page' => $vouchers->lastPage(),
            ],
        ]);
    }

    public function activate(Request $request, Voucher $voucher): JsonResponse
    {
        if ($voucher->status !== 'inactive') {
```

```
            return response()->json([
                'success' => false,
                'data' => null,
                'error' => ['code' => 'INVALID_STATUS', 'message' => 'Voucher tidak dalam status inactive'],
            ], 422);
        }

        $before = ['status' => $voucher->status];
        $voucher->update(['status' => 'active']);

        $this->logAdminAction(
            $request,
            AdminAction::ActivateVoucher->value, // tambah case di AdminAction
            'voucher',
            $voucher->id,
            $before,
            ['status' => 'active'],
        );

        return response()->json(['success' => true, 'message' => 'Voucher activated']);
    }
}
```

3. Enum action baru (`app/Enums/AdminAction.php`):

```
case ActivateVoucher = 'activate_voucher';
```

4. Test (`tests/Feature/Admin/AdminVoucherControllerTest.php`):

```
final class AdminVoucherControllerTest extends TestCase
{
    use RefreshDatabase;

    public function test_activate_requires_inactive_status(): void
    {
        $admin = User::factory()->create(['role' => 'admin']);
        $voucher = Voucher::factory()->create(['status' => 'active']);

        $this->actingAs($admin)
            ->postJson("/api/admin/vouchers/{$voucher->id}/activate")
            ->assertStatus(422)
            ->assertJsonPath('error.code', 'INVALID_STATUS');
    }
}
```

12.2 Checklist implementasi fitur admin baru

1. Tambah enum action baru ke `AdminAction` (jika aksi mutable)
2. Tambah route di group admin (`routes/api/admin.php`)
3. Buat controller dengan strict types + `LogsAdminActions`
4. Guard bisnis: return JSON 422 `{code, message}` — jangan throw exception tanpa handler

5. Tulis audit log (before/after snapshot) di setiap mutasi
6. Tambah halaman Inertia di `resources/js/pages/admin/**` + route `web.php`
7. Tulis test di `tests/Feature/Admin/` (factory user admin + `actingAs` + `assertJsonPath`)

---

13. Implementation Guidelines

13.1 Proses Implementasi Bertahap

Urutan membangun fitur admin baru: **enum action → model (state machine jika perlu) → route → controller → halaman Inertia → test**. Cross-cutting concern yang wajib diintegrasikan: role middleware (`role:admin`), audit log (`LogsAdminActions`), pola response `{success, data, pagination|meta}`.

13.2 Pitfall Umum

1. **Dua bentuk error guard** — pilih satu: return JSON 422 `{error: {code, message}}`. `InvalidCafeStatusTransitionException` saat ini menghasilkan 500; tambahkan `render()` di exception atau tangkap di controller untuk 422.
2. **Kolom legacy** `status='approved'` vs `approval_status` — selalu pakai `approval_status` (atau scope `Cafe::active()`), jangan `where('status', 'approved')` seperti `topCafes`.
3. **Refund & komisi tanpa guard** — `process` refund dapat menimpa refund completed; `updateRate` tanpa validasi range; tambahkan guard `INVALID_STATUS` dan validasi.
4. **`active_to` / `active_from`** pada `CommissionRule` & `FeaturedCafe` harus divalidasi relatif (`after:active_from`).
5. **N+1** — selalu eager-load relasi yang dipakai response (pola `with('owner', 'customer')` sudah konsisten).
6. **Perfomance analytics** — `commission()` melakukan `get()` lalu group di memory; untuk volume besar, gunakan `DATE_FORMAT` di query (group by DB).
7. **Suspension user tidak invalidasi token** — user yang di-suspend masih bisa memakai token Sanctum lama sampai `Sanctum::tokensCanExpire` / middleware cek `is_suspended` diterapkan.

13.3 Mekanisme Ekstensi

- **State machine** — tambah transisi baru cukup di `CafeApprovalStatus::TRANSITIONS` tanpa menyentuh controller.
- **Enum** `AdminAction` — satu sumber kebenaran untuk daftar aksi audit; `AdminDashboardController::activityMessage()` memetakan label (tambah label di map saat enum bertambah).
- **Rate komisi** — perubahan rate berlaku otomatis via `CommissionService::getActiveRate()` (date-range active), tanpa mengubah kode perhitungan.
- **Service DI** — injeksi `readonly` constructor; tambah service baru cukup dengan type-hint di constructor.

Kesimpulan

Pola yang paling penting untuk dipertahankan saat menambah fitur admin baru:

1. **AuthZ via middleware group** — `auth:sanctum` + `role:admin` di route group, bukan per-method.
2. **Controller tipis + service untuk logika domain** — `BookingService::cancel`, `CommissionService`; jangan taruh logika multi-langkah di controller.

3. **Audit di setiap mutasi** — `LogsAdminActions` dengan snapshot before/after + IP; konsisten gunakan enum `AdminAction`.
4. **State machine untuk status domain** — `transitionTo()` di model dengan transisi eksplisit, bukan update status bebas.
5. **Guard bisnis → 422 JSON terstruktur** — `{success: false, error: {code, message}}` (kecuali validasi form → `{errors}`).
6. **Response paginated konsisten** — `{success, data, pagination}` dan prefer eager-load + `per_page` dari request.
7. **Kompensasi dalam transaksi** — multi-tulis (suspend cafe + batal booking) harus satu `DB::transaction`.
8. **Test per controller** di `tests/Feature/Admin/` dengan factory + `actingAs` admin.